

EDUCATION

Veermata Jijabai Technological Institute (VJTI), Mumbai

June 2027

Bachelor in Technology in Electronics and Telecommunications.
Minor in Artificial Intelligence and Machine Learning

9.17 GPA - Present

TECHNICAL SKILLS

- **Programming languages:** C, C++, Python, SQL
- **Dev Tools:** Arduino IDE, ESP-IDF
- **ML and Data Analysis Toolset:** OpenCV, Numpy, Pandas, Excel
- **CAD Tools:** AutoCAD
- **Operating Systems:** Linux, Windows, MacOS

PROFESSIONAL EXPERIENCE

IIT Bombay, Mumbai, India | IoT Research Intern - Machine Intelligence Program

June 2026 - July 2026

- Independently conceived, pitched, and built ParkinSense, a wearable continuous monitoring platform for Parkinson's disease, from concept to a working end to end prototype (firmware, BLE protocol, real time signal processing, dashboard) during the internship.
- Extended the lab's existing nRF52840/BLE IMU expertise into a three pipeline architecture (motion, PPG/cardiac, and activity) fusing inertial and optical sensing on one synchronised 104 Hz stream, moving beyond the single modality tremor detection common in prior wearable PD research.
- Overcame key limitations of traditional approaches by replacing offline, single axis tremor analysis with real time, multi axis, motion context aware detection; replacing isolated activity tracking with a step counter cross gated against tremor and handling artifacts; and replacing single channel monitoring with a motion aware confidence model linking cardiac and activity readings back to tremor state.

Sciencetech, Dubai, UAE | Project Engineering Intern - Building Technologies Department

June 2025 - July 2025

- Gained hands-on exposure to the core components of a Professional AV system, including DSPs, matrix switchers, amplifiers, speakers, and control processors.
- Assisted in system design tailored to client requirements, considering use-case, venue acoustics, display sizing, and integration with lighting/BMS.
- Explored modern AV-over-IP solutions, understanding signal routing, network infrastructure, and system scalability for distributed AV environments.
- Designed initial I/O schedules (AI, DI, AO, DO) and conducted the DDC selection for several BMS projects (eg: Burj Binghatti Jacob & Co. Residences, GEMS Founders School, etc.). The selection process was guided by considerations of equipment schedules, load demand, spare margins, and peripheral device requirements.
- Implemented, tested, and commissioned a KNX-based lighting control system at Amazon DXB30 Innovation Hub, Dubai Internet City, including actuator configuration, addressing, and line diagnostics on a central monitoring system.

IIT Bombay, Mumbai, India | IoT Research Intern - Machine Intelligence Program

Feb 2025 - May 2025

- Conducted research on IoT-based sensor networks and real-time data transmission.
- Worked with embedded systems like nRF52840 Sense and ESP32 for high-speed IMU data acquisition and optimised BLE communication for efficient data transfer in IoT applications.
- Collaborated on an industrial project with V-Guard involving MPU data collection from underwater submerged pumps.
- Investigated frequent EMI-related disruptions in the existing serial communication setup using a waterproof cable.
- Contributed to transitioning the system from serial to a UDP-based wireless transmission approach, which improved data reliability and throughput in the presence of interference.
- Supported testing and validation of the wireless system to ensure consistent real-time data retrieval under operational conditions.

ACADEMIC PROJECTS

ParkinSense - Continuous Multi-Modal Wearable Platform for Parkinson's Monitoring (Embedded Systems, BLE, Signal Processing, Python)

- Architected a complete end to end wearable system, including custom firmware, BLE protocol, three parallel real time signal processing pipelines, and a live dashboard, for continuous Parkinson's disease monitoring, addressing the gap left by single modality tremor wearables that ignore cardiac and activity context.
- Built a 104 Hz sensor fusion pipeline on a Seeed XIAO nRF52840 with LSM6DS3 IMU and MAX30102 PPG sensor, streaming a versioned 248 byte binary BLE packet (10 samples per packet) to support simultaneous inertial and optical acquisition without throughput loss.
- Designed a motion pipeline (gravity removal, 50 Hz notch filtering, 4.0 to 7.5Hz Butterworth band pass, multi axis feature extraction) with a hysteresis based tremor state machine, replacing single axis/FFT only detection with multi axis scoring, temporal validation, and a decoupled confidence estimator to cut false positives from voluntary motion.
- Developed a PPG pipeline from scratch (finger detection with hysteresis, Signal Quality Index, adaptive peak detection, RR interval extraction) producing heart rate, HRV (RMSSD, SDNN, pNN50), and a ratio of ratios SpO₂ estimate using a SEARCHING to LOCKED to TRACKING signal lock state machine.

- Engineered a from scratch accelerometer based step tracking pipeline (adaptive mean+ $k\sigma$ threshold, hysteresis peak detection, refractory gating) explicitly gated by the motion pipeline's activity classifier, solving the false step problem where tremor or device handling gets miscounted as gait, a limitation unaddressed in standard pedometer algorithms.
- Built a decoupled live Plotly Dash dashboard and CSV/JSON logging layer, enabling offline replay and future ML dataset generation without coupling the UI to the analytics pipelines.

TREMOSense - Anti-Tremor Assistive Spoon for Parkinson's Patients (Embedded Systems, IMU Sensors)

- Engineered a 2-axis active stabilization system to suppress pathological hand tremors (4–12 Hz) while preserving intentional motion (<2 Hz), enabling improved usability for Parkinson's patients.
- Integrated LSM6DS3 6-DoF IMU with nRF52840 (ARM Cortex-M4F) to achieve real-time motion tracking at 100 Hz sampling rate (10 ms control loop).
- Designed and implemented a Multivariate Extended Kalman Filter (EKF) for accurate orientation estimation, eliminating gyroscope drift and high-frequency noise.
- Developed an optimal LQR (Linear Quadratic Regulator) controller for coordinated 2-axis stabilization, outperforming conventional PID by minimizing control effort and oscillations.
- Achieved low-latency closed-loop actuation using dual MG92B digital servos, generating precise PWM signals (550–2500 μ s) for real-time tremor compensation.
- Built a frequency discrimination pipeline to distinguish voluntary vs involuntary motion, ensuring natural user control during eating tasks.
- Optimized power electronics and transient handling, incorporating XL6009 boost converter + 1500 μ F decoupling capacitor to prevent voltage dips under peak servo loads (~1.5A).
- Validated system performance through Simulink and Simscape modeling, ensuring stable operation with <10° phase lag at ~6 Hz tremor frequency.
- Delivered a portable prototype (~4-hour battery life) with focus on lightweight design, usability, and real-world deployment.

NAVISIGHT (Arduino, TensorFlow Lite)

- Developed a comprehensive dual-component assistive system for the visually impaired: A **Smart Cane** with obstacle detection and an ESP32-CAM-based **Smart Vision Module** with real-time object recognition to enhance independent navigation, object awareness, and personal safety.

1. ECHO-STRIDE - Smart Cane for the Visually Impaired

- Engineered an innovative smart cane using **Ultrasonic sensors** and an **Arduino Uno** board for obstacle detection and real-time path awareness.
- Added haptic feedback through a vibration motor and buzzer, providing intuitive, real-time alerts.
- Designed a sleek, ergonomic casing to ensure durability and comfort, making it practical for daily, long-term use.
- Added **GPS** (L80-M39) and **Bluetooth** (HC-05) modules to periodically transmit the user's location to their smartphone.
- Configured mobile-side automation using Tasker to send the user's live Google Maps location to trusted contacts via SMS at set intervals.

2. EYECUE - ESP32-CAM Smart Vision Module

- Developed a wearable vision system using the **ESP32-CAM** for real-time image capture and object/person recognition.
- Trained and deployed lightweight TensorFlow Lite models on the ESP32-CAM through the Arduino IDE, enabling offline, low-power inference.
- Implemented image preprocessing and model inference directly on-device, with results (such as identified objects or persons) streamed to the serial monitor for debugging and development purposes.
- Currently working on integrating audio feedback to speak out the detected objects using TTS(Text-To-Speech), enhancing spatial awareness and scene understanding for the user.
- The module is being optimised for form factor and power efficiency, with the goal of creating a lightweight and practical wearable suitable for daily outdoor use.

Wall-E (ESP-IDF, GIT)

- Collectively designed and implemented a two-wheeled self-balancing and line-following robot named WALL-E, using an ESP32 microcontroller and a custom in-house PCB called the SRA Board (Designed by SRA)
- Integrated a Light Sensor Array (LSA) to detect surface colour based on light reflectivity differences, enabling the bot to follow white or black lines.
- Developed a PID control algorithm to process data from the LSA and adjust motor speeds for accurate line tracking.
- Created a self-balancing system that uses positional feedback and PID control to maintain stability on two wheels during movement.
- Faced challenges in tuning PID parameters, requiring iterative testing and adjustments to achieve desired performance. The robot demonstrated reliable performance in both line-following and self-balancing tasks in various test scenarios.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Academic Branch Topper (First Year)
- Was part of organising Asia's second-largest TechFest, Technovanza, with a footfall of 4,000-5,000 participants (2024).
- Member of SRA (Society of Robotics and Automation), VJTI, actively participating in workshops.
- Played guitar at various events, contributing to various cultural programs and in musical performances.